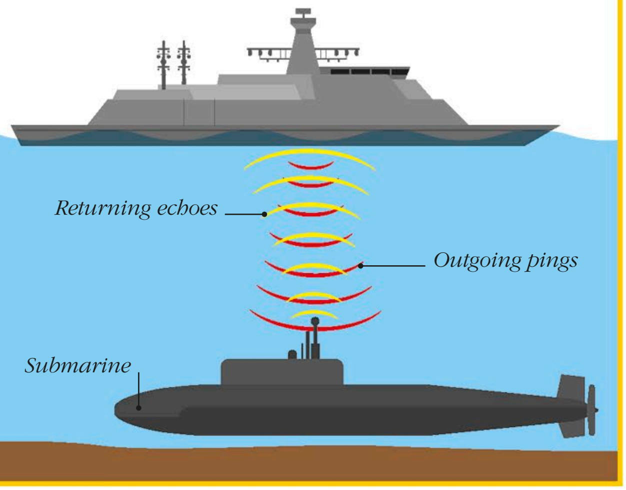
KEEPING PERFECT TIME

To find out their longitude—how far east or west they had sailed—navigators needed to check at midday what the exact time was back home. The time difference showed the distance. However, ordinary pendulum clocks lose time on a rocking ship. In 1735, English clock maker John Harrison made a marine chronometer, which kept perfect time on the move.

WOW! Thanks to electronic satellite systems, some ships can navigate around the world automatically.

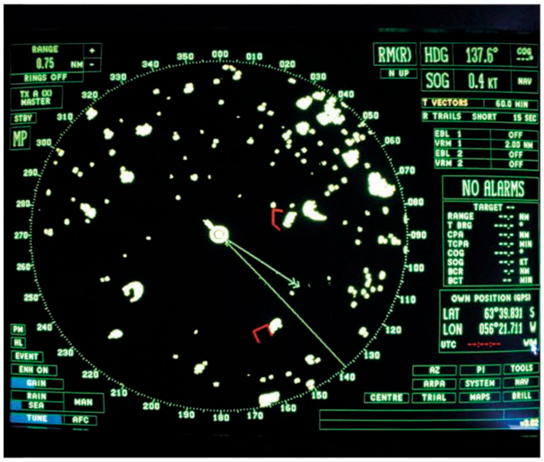
SONAR

This system uses sound waves to detect objects under the sea. It sends out "pings" and antennas pick up the ways they bounce back or echo. The American naval architect Lewis Nixon invented the first sonarlike listening device in 1906 to detect icebergs. In 1915, during World War I, the French physicist Paul Langévin built the first functional sonar to detect submarines.



RADAR

In 1904, radar systems were pioneered by the German inventor Christian Hülsmeyer. He realized he could reveal hidden objects, such as ships shrouded by dense fog, by bouncing radio waves off them. These resulted in echoes, which made a bell ring or, in later versions, made a glowing spot on a screen. During World War II, radar was used to detect enemy ships. Since then, radars have become key navigation aids.



Radar screen of a research ship shows a field of icebergs